



Lab 02: Introduction to SQL for BigQuery and Cloud SQL

Develop Your Google Cloud Network | GSP281

Overview

Learned SQL fundamentals using BigQuery on the London bikeshare public dataset. Exported data to CSV files, uploaded to Cloud Storage, and used Cloud SQL to create databases, tables, and run advanced queries.

Objectives

- Run SQL queries in BigQuery on a public dataset
- Use SELECT, FROM, WHERE, GROUP BY, COUNT, AS, ORDER BY
- Export query results to CSV and upload to Cloud Storage
- Create a Cloud SQL instance, database, and tables
- Load CSV data into Cloud SQL tables
- Run advanced queries using DELETE, INSERT INTO, and UNION

Key SQL Keywords

Keyword	Purpose
SELECT	Specify which fields to return
FROM	Specify which table to query
WHERE	Filter rows by a condition
GROUP BY	Aggregate rows that share the same value
COUNT	Count the number of rows matching criteria
ORDER BY	Sort results ascending or descending
UNION	Combine output of two or more SELECT queries

Task 1 - BigQuery Queries

Queried the London bikeshare public dataset using BigQuery to find trips over 20 minutes and top starting stations.

```
SELECT end_station_name FROM `bigquery-public-data.london_bicycles.cycle_hire`;  
SELECT * FROM `bigquery-public-data.london_bicycles.cycle_hire` WHERE  
duration>=1200;  
SELECT start_station_name, COUNT(*) AS num FROM `bigquery-public-  
data.london_bicycles.cycle_hire` GROUP BY start_station_name ORDER BY num DESC;
```

Query results

Chat Save results Open in

Job information Results Visualization JSON Execution details

Row	end_station_name
1	Ontario Street, Elephant & Castle
2	York Way, Kings Cross
3	Euston Road, Euston
4	Euston Road, Euston
5	New Cavendish Street, Maryle...
6	Newgate Street, St. Paul's
7	Abingdon Green, Westminster
8	Abingdon Green, Westminster
9	Sea Containers, South Bank
10	Sea Containers, South Bank

Results per page: 50 1 - 50 of 83434866

Task 2 - Export to Cloud Storage

Exported query results as CSV files and uploaded to a new Cloud Storage bucket.

- Exported start_station_name.csv and end_station_name.csv
- Created a Cloud Storage bucket and uploaded both CSVs

Bucket details

Edit bucket Go to path Refresh

sql_bucket_niyah

Location: us (multiple regions in United States) Storage class: Standard Public access: Not public Protection: Soft Delete

Objects Configuration Permissions Protection Lifecycle Obs

Create folder Upload Transfer data

Name	Size	Type	Created
end_station_name.csv	34.6 KB	text/csv	Jun 10, 2
start_station_name.csv	9.1 MB	text/csv	Jun 10, 2

1 file successfully uploaded

Uploads and qwiklabs-gcp-04-4178... operations

- end_station_name.csv Complete
- start_station_name.csv Complete

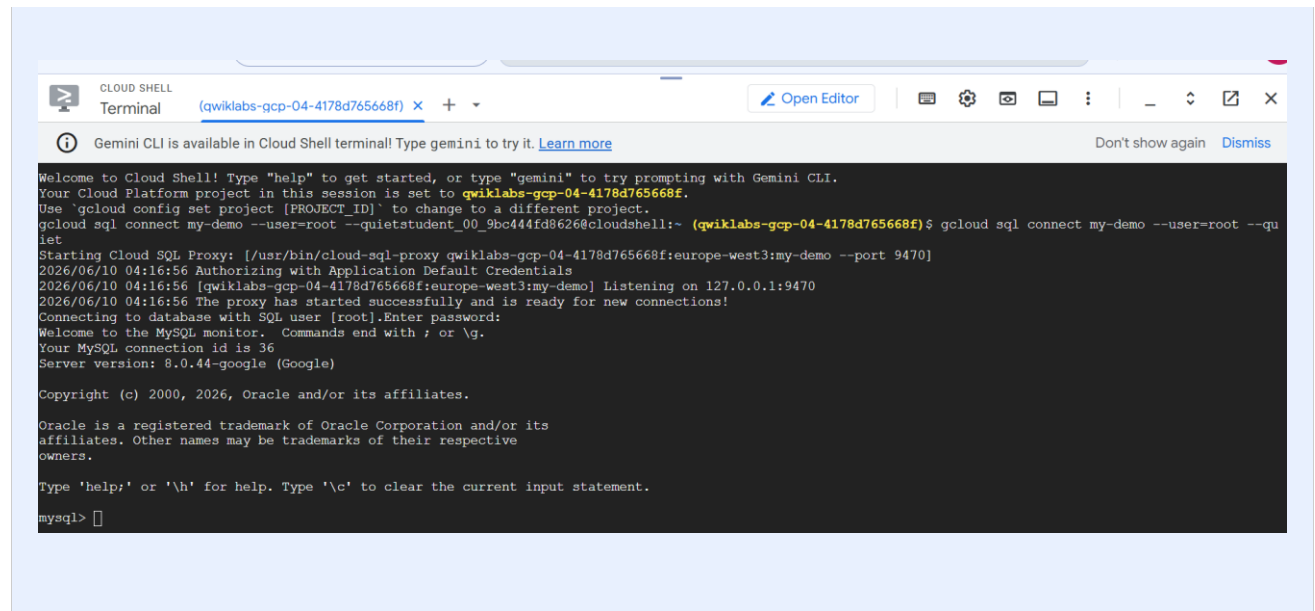
Get started with Cloud Storage

- Getting bucket information
 - Help document
 - Get information on the size and metadata of your Cloud Storage buckets.
- Uploading objects
 - Help document
 - Upload the objects containing your data to your Cloud Storage buckets.
- Downloading objects
 - Help document
 - Download the objects from your Cloud Storage buckets.
- Use cases for Cloud Storage
 - Help document

Task 3 - Cloud SQL Instance

Created a MySQL Cloud SQL instance and connected via Cloud Shell.

```
gcloud sql connect my-demo --user=root --quiet
CREATE DATABASE bike;
USE bike; CREATE TABLE london1 (start_station_name VARCHAR(255), num INT);
USE bike; CREATE TABLE london2 (end_station_name VARCHAR(255), num INT);
```



Task 4 - Advanced Queries

Used DELETE, INSERT INTO, and UNION to manipulate and combine data across tables.

```
DELETE FROM london1 WHERE num=0;
INSERT INTO london1 (start_station_name, num) VALUES ("test destination", 1);
SELECT start_station_name AS top_stations, num FROM london1 WHERE num>100000 UNION
SELECT end_station_name, num FROM london2 WHERE num>100000 ORDER BY top_stations
DESC;
```

```

| Bethnal Green Road, Shoreditch | 322209 |
| Berry Street, Clerkenwell | 178864 |
| Bermondsey Street, Bermondsey | 178368 |
| Belvedere Road 1, South Bank | 110618 |
| Bell Lane, Liverpool Street | 108888 |
| Belgrove Street , King's Cross | 586568 |
| Belgrave Road, Victoria | 134847 |
| Beaumont Street, Marylebone | 103035 |
| Bayswater Road, Hyde Park | 223381 |
| Baylis Road, Waterloo | 223294 |
| Bayley Street , Bloomsbury | 155304 |
| Battersea Power Station, Battersea Park | 149642 |
| Bath Street, St. Luke's | 163819 |
| Barons Court Station, West Kensington | 108586 |
| Bankside Mix, Bankside | 201623 |
| Bank of England Museum, Bank | 186018 |
| Baldwin Street, St. Luke's | 110713 |
| Ashley Place, Victoria | 183860 |
| Aquatic Centre, Queen Elizabeth Olympic Park | 309576 |
| Ansell House, Stepney | 103641 |
| Allington Street, Victoria | 129176 |
| All Saints Church, Portobello | 137984 |
| Aldersgate Street, Barbican | 190311 |
| Alderney Street, Pimlico | 120167 |
| Albert Gate, Hyde Park | 464713 |
| Albert Bridge Road, Battersea Park | 127936 |
| Ada Street, Hackney Central | 111076 |
| Abingdon Green, Westminster | 159305 |
| Abbey Orchard Street, Westminster | 145521 |
+-----+
313 rows in set (0.28 sec)

mysql> 

```

What I Learned

- BigQuery can query petabyte-scale datasets with standard SQL
- GROUP BY returns unique values and COUNT shows how many rows match
- Cloud SQL is a managed relational database that accepts CSV imports
- UNION combines results from multiple SELECT queries into one output
- DELETE removes rows matching a condition, not just the first row

Resources

BigQuery Docs: <https://cloud.google.com/bigquery/docs> | Lab ID: GSP281